# Explainable Fake Job Advertisement Detection Framework

## A Trust-Aware, Auditable Decision Support System for Online Recruitment Fraud

**MODEL BASELINE:** Fraud-BERT (Discover Computing, 2025)

**CORE STACK:** React / Node.js / FastAPI / MongoDB

**INTEGRATION:** SHAP & LIME XAI, Heuristics Scorecard, OCR Ingestion

# 1. Introduction & Motivation

## Problem Statement

- **Explosion of Recruitment Fraud:** The rapid growth of online job hunting has made job seekers vulnerable to sophisticated spoofing schemes and fake postings.
- **Significant Candidate Harm:** Leads to severe outcomes including financial extortion (fees) and harvesting of personal identification info.
- **Opaque Black-Box Models:** Modern classifiers predict fraud probabilities but offer zero explanation of results, leaving candidates unable to verify details.

## Proposed Project Vision

- **Actionable Interpretability:** Move from opaque binary classifications to detailed warnings showing exactly why a post is flagged as suspicious.
- **Domain-Specific Heuristics:** Cross-validate model evaluations against rule-based indicators (e.g. registration fee requests, free domains).
- **Auditable Trust Decisions:** Empower candidate decision support with dynamic explanation panels (SHAP / LIME) and risk badges.
- **Ingestion Adaptability:** Provide uniform scanning pipelines for structured URLs, raw text, documents (PDF), and screenshot images (OCR).

# 2. Base Paper vs. Proposed Framework

| Feature | Fraud-BERT Baseline (Taneja et al., 2025) | Proposed Extension Framework |
|---|---|---|
| **ML Model Core** | Standard BERT-base-uncased classifier. | BERT-base-uncased with tuned decision boundaries. |
| **Interpretability** | None (Black Box). Labels only. | Interactive SHAP (Global population features) and LIME (Local word explanations). |
| **Trust Indicators** | None. | 0-100 deterministic Trust Score based on 8 heuristics (domain, logo, contact checks). |
| **Severity Index** | None. | 0-100 danger indicator detecting money requests and WhatsApp/Telegram recruitments. |
| **Risk Tiering** | None. | 4-tier Risk Badge (Low, Medium, High, Critical) mapping BERT prob and Trust Score. |
| **Ingestion Modes** | Text fields only. | Supports URL scraping, PDF extraction, screenshot image OCR, and raw text. |

# 3. System Architecture Design

## Frontend (React 19)

- Ingests job links, raw text, PDF files, and image files.
- Displays the interactive Explainability Dashboard.
- Renders LIME highlighting and SHAP feature panels.
- Connects only to Node.js backend (strict separation).

## Backend (Node/Express)

- Authentication & session management.
- Enforces rate limiting and input validation rules.
- Logs query details and transaction results to MongoDB.
- Proxies verified requests to Python FastAPI AI service.

## AI Service (FastAPI)

- Executes tokenization and clean preprocessing.
- Fine-tuned BERT classification inference.
- Calculates Trust, Severity and Risk Score metrics.
- Precomputes SHAP cache; runs LIME iterations on request.

## Decoupled Data Architecture & Deployment Setup

- Database Layer: MongoDB houses 'users' (auth details) and 'analyses' schemas (saving predictions and audit records).
- Performance Caching: The BERT model classifier and the precomputed global SHAP values are loaded once on FastAPI startup to avoid local execution bottlenecks.

# 4. Dataset & Data Imbalance Insights

## EMSCAD Dataset Properties

- **Total Records:** 17,880 postings mapped across 18 schema variables.
- **Severe Imbalance:** 17,014 Legitimate postings vs. only 866 Fraudulent postings (4.84%).
- **weighted loss:** To prevent classifier collapse (where models optimize simple accuracy by predicting only the majority class), we apply class weights in our cross-entropy loss function during BERT training.
- **Text Preprocessing:** Strips HTML tags, normalizes Unicode characters, aggregates nested fields (profile, requirements), and performs WordPiece tokenization.

## Missingness Percentages: Legit vs Fraud

| Field Name | Missing (Legit) | Missing (Fraud) |
|---|---|---|
| company_profile | 15.99% | 67.78% |
| salary_range | 84.45% | 74.25% |
| benefits | 40.30% | 42.03% |
| requirements | 14.95% | 17.78% |

### Signal Validation Insight:

The significant difference in missing company profiles (67.78% in fraud vs. 15.99% in legitimate postings) confirms that lacking profile details is a strong warning sign. This directly validates the rule penalties used in the trust score scorecard.

# 5. AI Classification Pipeline Steps

## 1. Input Ingest

- Extracts text from scraped URLs, processes PDF files using PyMuPDF, or runs EasyOCR on screenshot images.

## 2. Preprocess

- Strips HTML characters, cleans emojis, normalizes whitespace formats, and generates WordPiece tokens.

## 3. ML Inference

- Fine-tuned BERT-base-uncased computes raw classification probability. Applies tuned 0.65 threshold.

## 4. Output Engines

- Runs heuristic calculations for Trust/Severity scorecards and runs live LIME explainability models.

# 6. Explainable AI (XAI) Integration

## LIME (Local/Instance Explanations)

- **Real-time Perturbation:** Runs live per evaluation request, perturbing input sentences locally to observe model classification score fluctuations.
- **Word-Level Highlighting:** Identifies specific words driving predictions, color-coding features for job seekers (Legitimate signals vs. Fraud signals).
- **Transparency Focus:** Highlights suspicious terms like 'urgent', 'registration fee', and 'wire transfer' so applicants know exactly why a warning was raised.

## SHAP (Global/Population Explanations)

- **Population Analysis:** Computes cooperative game theory attribution scores across the entire EMSCAD training dataset.
- **Startup Optimization:** SHAP values are precomputed offline due to high overhead, cached on disk, and loaded immediately at application start.
- **System Verification:** Provides developers with an overview of what the BERT model learns overall, ensuring it avoids dataset bias or noise.

# 7. Deterministic Trust Score Engine

## Transparent Trust Evaluation

- **Anti-Stacking Approach:** Avoids placing a second opaque ML model on top of BERT. Instead, standard deterministic rules are used to score trust.
- **Heuristic Deductions:** Starts at 100 points. Penalties are deducted when specific anomalies are detected in text or metadata.
- **8 Active Indicators:** Checks contacts for free email domains (Gmail/Yahoo), missing company logo, registration fee mentions, missing location coordinates, and layout anomalies.
- **Layout Adaptability:** Identifies if text comes from unstructured formats (PDF/OCR) to avoid penalizing missing field structures.

## Trust Scoring Bounds

- **90 - 100:** High Trust: Postings from verified recruiters with complete company profiles and zero warnings.
- **70 - 89:** Moderate Trust: Minor missing parameters with valid contact info and standard text layout.
- **50 - 69:** Low Trust: Multiple warning signals detected. High missingness rates. Job seeker caution advised.
- **Below 50:** Unreliable / Highly Suspicious: Severe warning triggers (e.g. fees requested). Extreme missingness.

# 8. Risk & Severity Scorecards

## Fraud Severity Score (0-100)

- **Impact Metric:** Evaluates the harm a fraudulent listing poses to candidate safety, rather than just raw prediction likelihood.
- **6 Severe Signals:** Detects payment checks, bank account setup forms, wire transfer requests, and application processing fee rules.
- **Communication Risks:** Deducts major points if recruitment is directed strictly to mobile messaging apps (WhatsApp, Telegram, SMS).
- **Classification Safeguard:** High severity triggers elevate final risk recommendations, even if the ML classifier probability is low.

## Risk Categorization Matrix

- **Bivariate Mapping:** Maps coordinates of the BERT raw probability and the Heuristic Trust Score to output 4 discrete risk tiers.
- **Risk Badges:** Low Risk (Green), Medium Risk (Amber), High Risk (Orange), and Critical Risk (Red).
- **Probability Mitigation:** If BERT classifier output is low (e.g. 0.15) but the trust score is poor, the risk category is escalated to Medium.
- **Audit Trails:** Every risk categorization is logged in MongoDB to support developer audit reviews.

# 9. Model Results & Performance

## Metrics on Held-Out Test Set (2682 records)

- **Accuracy Score:** 98.73% classification accuracy on unseen EMSCAD test instances.
- **Precision Metric:** 90.68% - high precision reduces false positive warnings for legitimate recruiters.
- **Recall Metric:** 82.31% - leaves room for optimization via focal loss adjustments.
- **F1-Score:** 0.863 overall F1-score in detecting fraudulent listings.
- **ROC-AUC Score:** 0.983 area under curve shows strong discrimination performance.

## Test Set Confusion Matrix

| | Pred Legitimate | Pred Fraudulent |
|---|---|---|
| **Actual Legitimate** | **2,541 (TN)** | **11 (FP)** |
| **Actual Fraudulent** | **23 (FN)** | **107 (TP)** |

### Baseline Validation Summary:

Accuracy (98.73%) is very close to the base paper baseline (99%). Our recall (82.31%) is slightly lower than baseline (93%) in this run. Precision (90.68%) is highly optimized to avoid warning user candidates unnecessarily.

# 10. Key Contribution & Future Scope

## Research Novelty

- **Dual XAI Integration:** Combines local explainability (LIME) and global explainability (SHAP) with deep BERT classification.
- **Bivariate Risk Framework:** Combines neural network probabilities with deterministic heuristics to compute a reliable scorecard.
- **Multi-Input Modality:** Supports scraped URLs, parsed PDF files, raw text, and screenshot images (OCR pipelines).
- **Auditable Codebase:** Comprehensive integration tests verify classifier outputs across edits.

## Future Research Roadmap

- **Browser extension:** Real-time in-page scanning tool to analyze listings on LinkedIn and recruitment boards.
- **Multilingual Support:** Add tokenizers and classifiers for listings in Spanish, German, French, etc.
- **Continual Learning:** Deploy pipelines to automatically update model weights as new scams are reported.
- **Platform APIs:** Enable recruitment portals to query listings via validation APIs.

# Conclusion & Q&A

A major step forward in recruitment security, bringing transparency to fake job classification.

## Repository Blueprint Summary

- ai_service/ - Python FastAPI model processing logic.
- backend/ - Node.js MVC Gateway & MongoDB connection layer.
- frontend/ - React Dashboard with interactive LIME visualizations.
- docs/ARCHITECTURE.md - Full three-tier architecture definitions.

## Questions & Discussion

We welcome your questions regarding model training, SHAP/LIME integration thresholds, or scoring calculations.